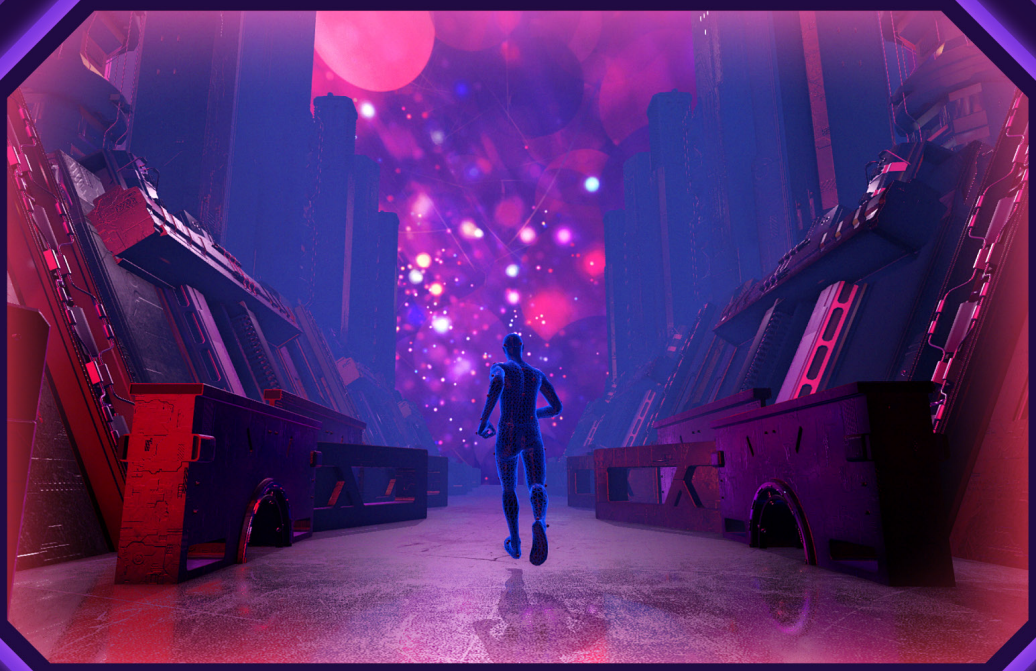


INTO THE METAVERSE

A COMPREHENSIVE REPORT

chaindebrief.

01 June 2022

PREFACE

This report is designed conceptually for beginners to increase their understanding of the diversity of NFTs and the current state of the Metaverse. More complex concepts are not covered in this report.

The report is broken up into several bite-sized sections to be digested easily.

Source and credit references are in the notes.

References not inside notes:

1. Grayscale November 2021
2. The Block Digital Asset 2022
3. NewZoo Trend Report Free 2021
4. Goldman Sachs Framing the Future
5. OV Metaverse Report 2021
6. Morgan Stanley Metaverse Research Report

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1. WHAT IS THE METAVERSE

1.1 What is the Metaverse
1.2 Understanding the Metaverse
1.3 Metaverse platforms
1.4 Key Metaverse Elements
1.5 Estimated Cost of the Metaverse

1. WHAT IS THE METAVERSE

1.1 - WHAT IS THE METAVERSE?

The internet is in the midst of transitioning from Web2.0 to Web3.0, shifting investor perception towards innovation and decentralization. With technology evolving, **Web3.0 aims to go beyond simply connecting people.**

The Metaverse is seen as an early framework for this "new internet". It is intangible and undefined. Various reports speculate on the possibilities, but the only undisputed conclusion is that it will be a better internet for everyone.

Firstly, a key component of the Metaverse would be VR (Virtual Reality), characterized by persistent virtual worlds

It also refers to an interoperable digital decentralized economy, allowing users to purchase virtual items. A Metaverse also allows you to **create a persona** similar to popular social media platforms like Instagram, Tiktok, or Facebook.

It's difficult to parse this **because these structures already exist.** What exactly is different this time? Some argue that the key factor of the Metaverse is doing away with centralized entities, with many companies working on Metaverse infrastructure. This includes Meta, Google, Nvidia, and more. The thing is, **none of these companies actually know** what it is. Meta argues for fake houses to hang out in, while Microsoft thinks it involves virtual meeting rooms for new hires. While there is a rough framework for certain aspects of it, it is still too early to speak for this intangible concept and is instead up to us to decide.

Sources:

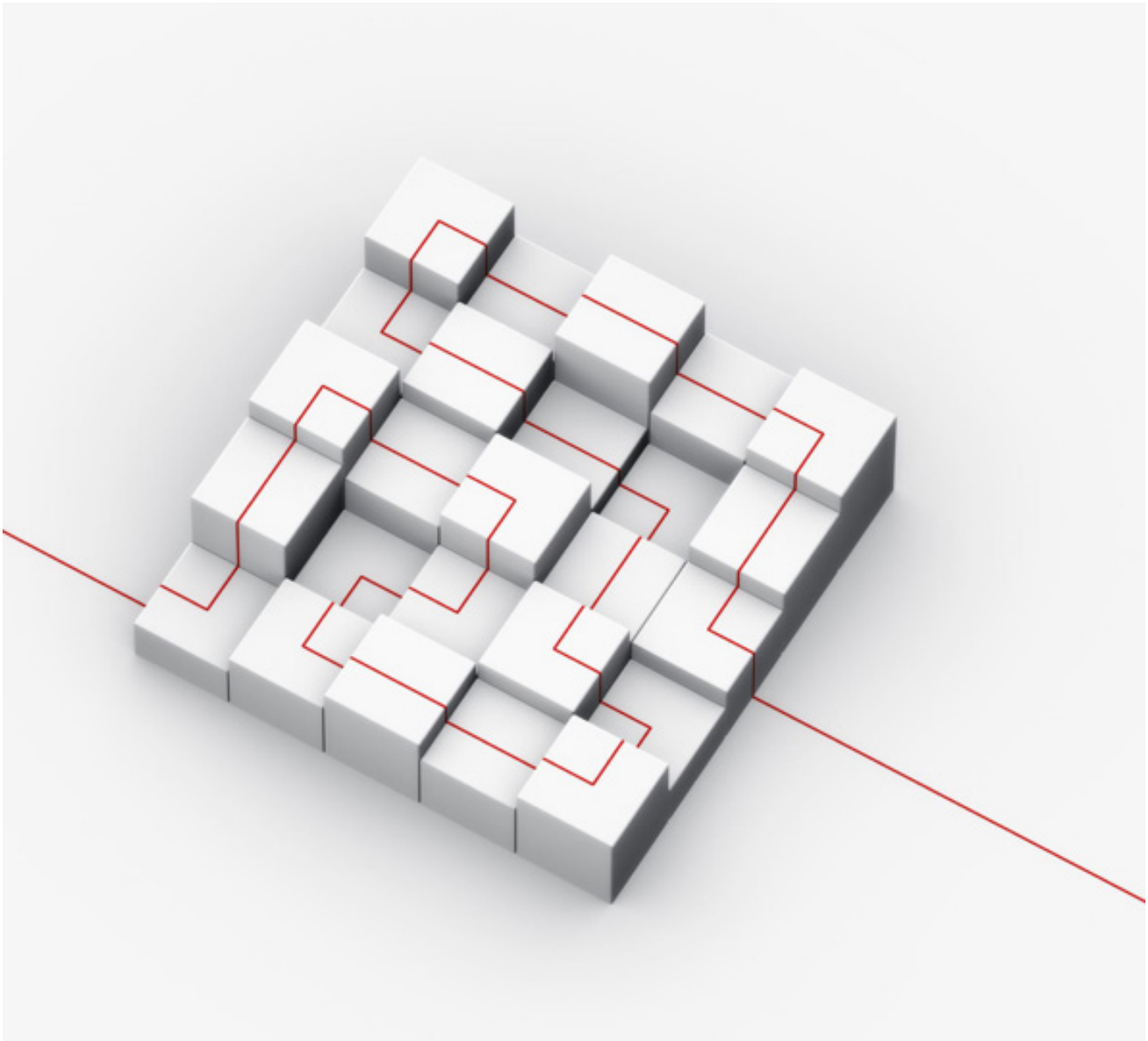
1. https://www.wired.com/story/what-is-the-metaverse/

1. WHAT IS THE METAVERSE

1.2 - UNDERSTANDING THE METAVERSE

The Metaverse also is built from the infrastructure of Web 3.0, the new internet born out of decentralized architecture. Interestingly, Web 3.0 is also an intangible concept but most would argue that it is quite well defined with some slight modifications.

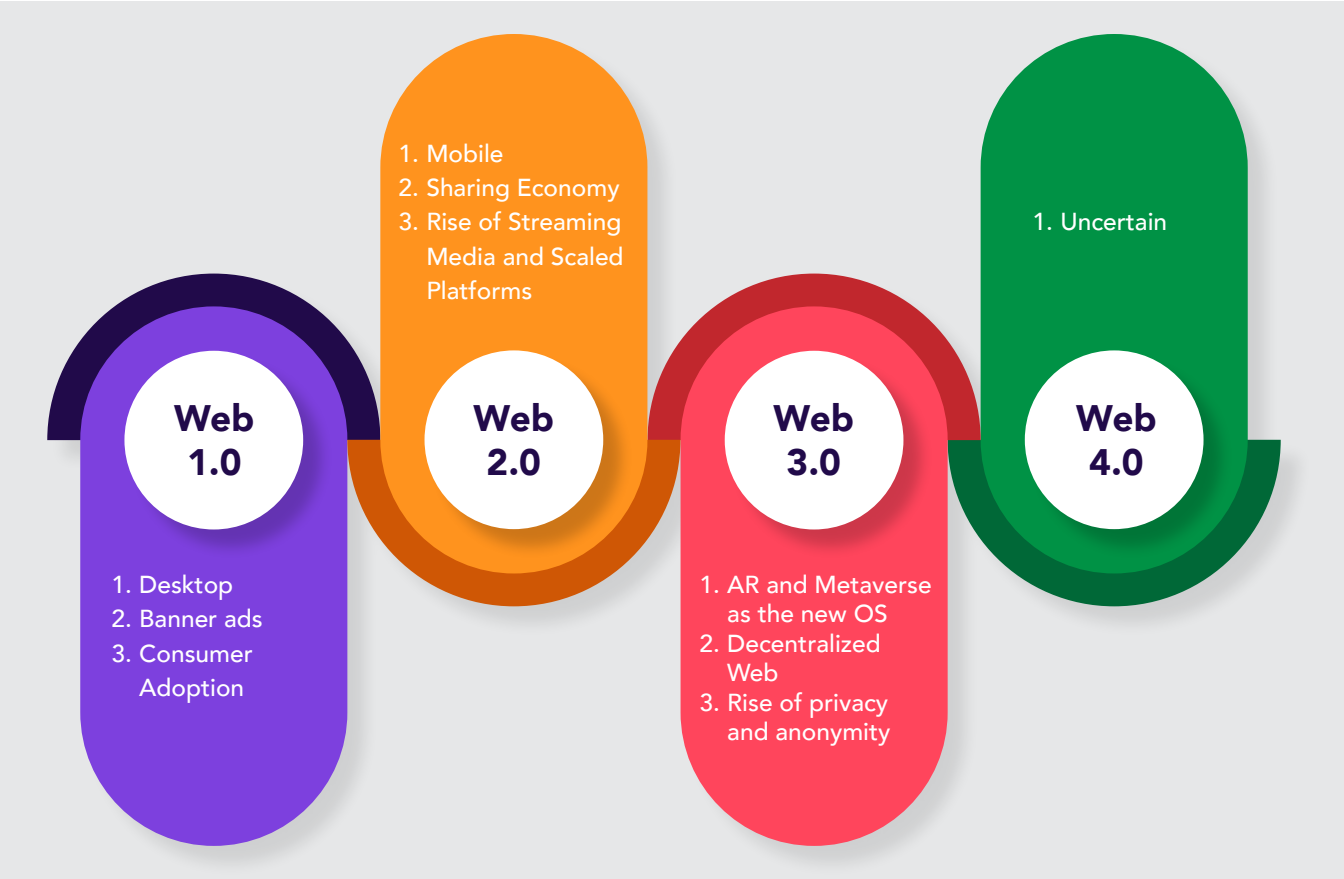
Our internet has evolved rapidly across eras, from Web 1.0 to Web 3.0. Web 2.0 was seen as the scale of users interacting with communities via mobile and desktop applications and services through a centralized entity. The defining characteristics of Web 3.0 are completely different, dabbling areas like decentralization, industry input costs, potential shift in media trends and more.



1. WHAT IS THE METAVERSE

1.2 - UNDERSTANDING THE METAVERSE

	Web 1.0	Web 2.0	Web 3.0
Interact	Read	Read Write	Read Write Own
Medium	Static Text	Interactive Content	Virtual Economies
Organization	Companies	Platforms	Networks
Infrastructure	Computers	Cloud & Mobile	Blockchain Network
Control	Decentralized	Centralized	Decentralized (or possibly a hybrid)



1. WHAT IS THE METAVERSE

1.3 - METAVERSE PLATFORMS

Interestingly and as expected, Metaverse Platforms all have different views. This gives rise to the Meta Market Opportunity. (more on that later)

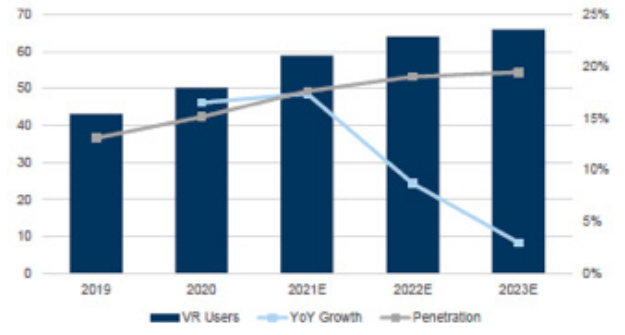
The CEO of Meta, Mark Zuckerberg, thinks that Metaverse as the successor of the internet will:

1. Elevate physical world experiences – Mixed reality of both the physical and real-world will magnify the feeling or presence. There are already many use cases for AR, which could serve as a template for businesses to utilize AR in the Metaverse
2. Co-created & built responsibility – What the Metaverse is defined as will come out in the next decade. In the near term, consumers will experience the Metaverse through 2D apps. Zuckerberg also believes that **many entities will collaborate** to build the Metaverse with privacy, integrity and safety.

The CEO of Roblox, David Baszucki, argues that the Metaverse is a virtual universe, persistent and shared, where platforms connect people from different realms of life, enabling communication in a new way. He argues that Roblox could be a medium for the Metaverse as a human co-experience

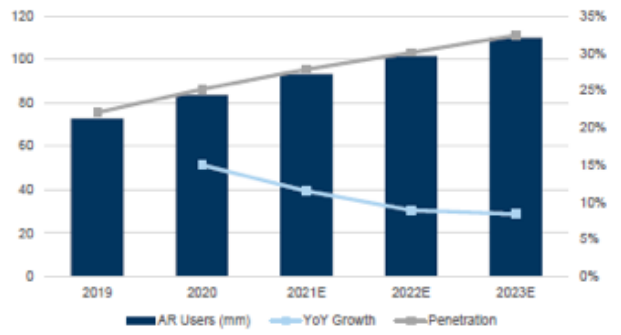
The CEO of Epic Games, Tim Sweeney, envisions the Metaverse as an expansive communal and virtual world, allowing users to interact with brands, intellectual properties, and many more. He believes individuals will build the Metaverse via user-generated content and there will be a free and fair economy where users can partake, make money, and be rewarded. Any users from major developers to individuals are all on equal ground.

Exhibit 4: US Virtual Reality Users
(mm, 2019-2023E)



Source: eMarketer, Data compiled by Goldman Sachs Global Investment Research

Exhibit 3: US Augmented Reality Users
(mm, 2019-2023E)



Source: eMarketer, Data compiled by Goldman Sachs Global Investment Research

It is estimated that technological advancements like 5G and edge computing with Metaverse use-cases will speed up adoption. However, low penetration rates in the next two years indicate Metaverse adoption is still early.

1. WHAT IS THE METAVERSE

1.4 - KEY METAVERSE ELEMENTS

The Metaverse is more than just a virtual reality. It is predicted that it will incorporate multiple key elements across 2 main sectors:

Gaming

1. New Social modes of in-game behavior and engagement
2. Rising popularity in NFTs
3. A rise in gaming guilds
4. Merging of game worlds and IPs
5. A blend of digital and physical aspects

Improving metrics across acquisition, retention, and monetization

1. Adding new revenue streams and finding new business models
2. Expanding reach to new audiences including non-gamers
3. Customer acquisition with new reasons and motivations to play

1.5 - ESTIMATED COST OF THE METAVERSE

1. Data Center Infrastructure
   - In the last decade, Meta has invested $17 billion in 14 data centers in the US and $4.1 billion internationally. They plan to spend much more with 70 additional buildings in the coming years to support the Metaverse

2. Further iterations of VR and AR products
   - Meta acquired Oculus for $2 billion in 2014 and has since spent a cumulative total of $2.5 billion. It plans to invest another $3 billion in VR products. Metaverse has accelerated its investments recently, purchasing 6 VR studios

3. Hiring Talent
   - 1/6 of Meta's employees focus solely on the Metaverse. They plan to double this in the next 5 years. Hiring specialized talent is extremely expensive and many of these employees rake 6-7 digits salaries annually.

4. Content Creators
   - Content Creators are essential for the Metaverse also. Meta has announced a $150 million fund to train creators with Metaverse skillsets.

There are many other public companies investing in the Metaverse. These include Amazon, Microsoft, Tesla, Apple, Google, and many more.

In 2021, $10.4 billion in capital was raised by these companies in 612 deals. It is estimated that these deals will have an ROI ranging from $135 billion to $1.35 trillion in the next three years.

2. HISTORY AND EVOLUTION

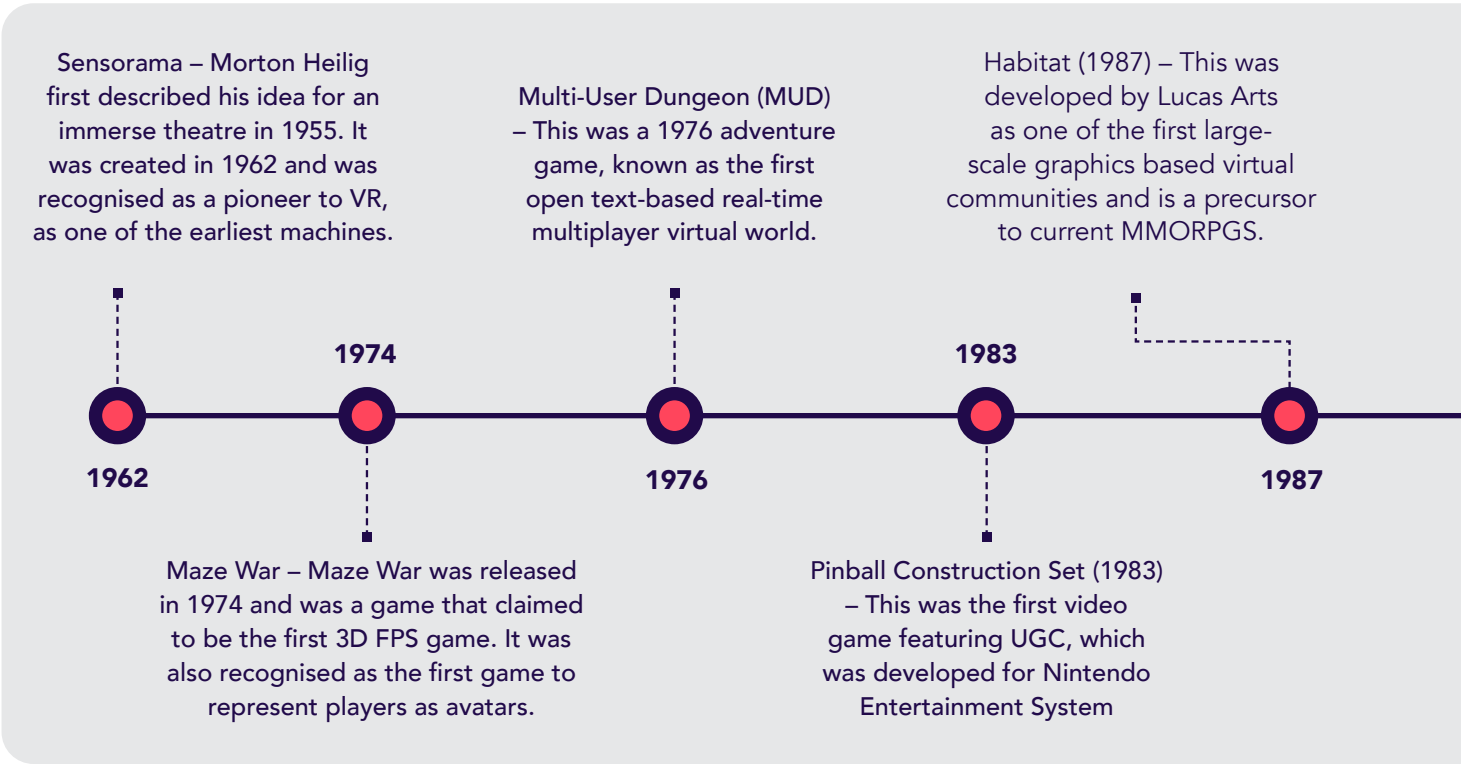
2.1 History of Virtual Worlds and Economies
2.2 Industries in the Metaverse
2.3 How the Metaverse is relevant to GameFi
2.4 Metaverse expansion: HSBC Case Study
2.5 Key themes of the Metaverse
2.6 Why is a Digital Identity important?

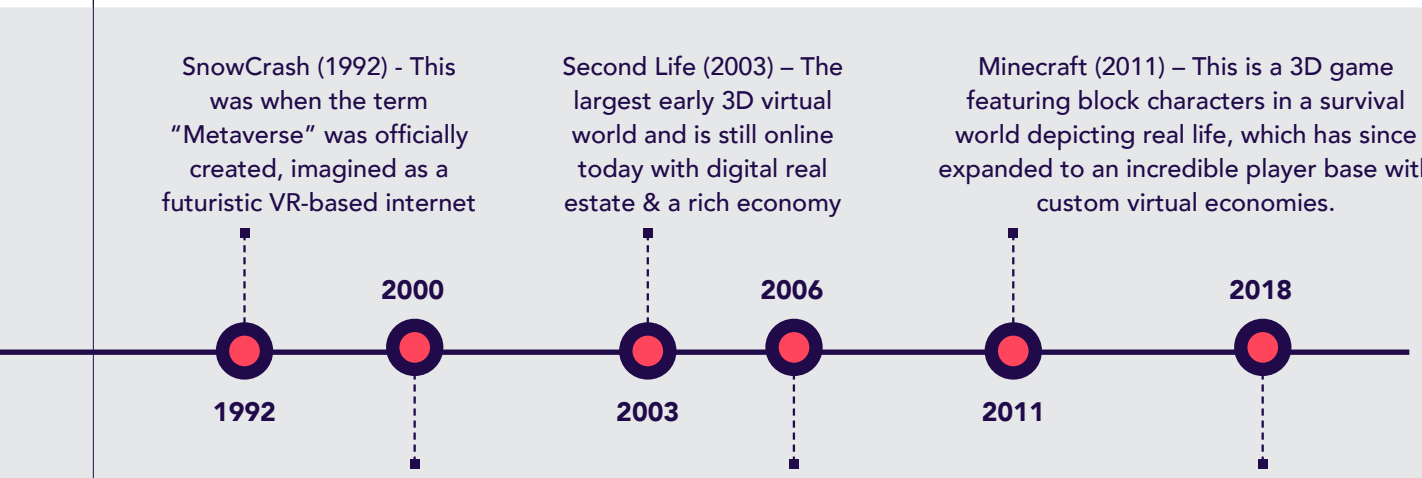
2. HISTORY AND EVOLUTION

2.1 - HISTORY OF VIRTUAL WORLDS AND ECONOMIES

Interestingly, the Metaverse is not a new concept. It has just evolved over time, representing the virtual world perspectives of different generations.

It has significantly advanced from multi-sensory machines, recognized as early virtual reality, to 3D virtual worlds with virtual economies. It is only expected to grow and mature, maybe even develop as an industry classification itself in the coming decade.





2. HISTORY AND EVOLUTION

2.2 - INDUSTRIES IN THE METAVERSE

The Metaverse makes use of blockchains, and ultimately smart contracts, so it is highly likely that every industry is likely to have a form of business opportunity or relevance in the future.

Currently, early activities have centered around brand marketing by collaborating with games to give players a way to express themselves or engage with their favorite content. In return, brands receive exposure, audience expansion, and better or newer revenue stream opportunities.

There are also several enterprise applications that will utilize games or game technology as a platform. Currently, the Metaverse is seen in:

Currently, the Metaverse is seen in

Music . TV & Film . Fashion and Cosmetics . Sports . Education & Training . Art . Automotive . Tourism . Retail . Offices . Factories

It is expected to be seen in

Transportation . Defence . Cities . Medicine



2. HISTORY AND EVOLUTION

2.3 - HOW THE METAVERSE IS RELEVANT TO GAMEFI

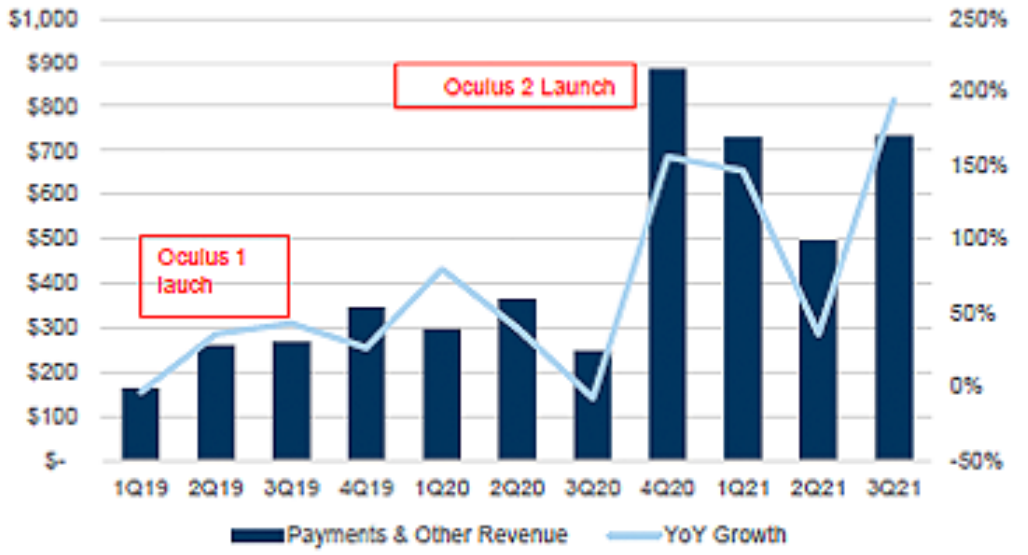
The Metaverse creates new modes of engagement with players and non-players, utilizing Web 3.0 infrastructure and decentralization.

It is expected to fuel the next big shift from Games-as-a-Service to Games-as-a-Platform, evolving gaming over the next decade to become an experience with amazing engagement.

Prior to COVID-19, games were already blurring the lines between virtual and physical reality, with "physical" in-game events in popular games like Fortnite and Roblox. These key events have increased user acquisition and engagement, leading to more conversions.

For example, Meta has experienced significant revenue growth with the launch of Metaverse products such as the Oculus 1 and 2.

Exhibit 7: Meta Platforms Payments & Other Revenue
$mm, Q1 '19 - Q3 '21



Source: Company data, Goldman Sachs Global Investment Research

The Metaverse is now at a prime stage of integrating non-gaming experiences, driven by technology and consumer interaction with games, allowing multiple stakeholders to create and capture value beyond its core product.

Publishers are also driving key experiences through other experiences such as virtual fashion shows, concerts, and media partnerships. Players now use the Metaverse to showcase their experiences and social identity. This highlights that while the Metaverse has grown GameFi significantly, its use cases have also expanded to various other industries.

2. HISTORY AND EVOLUTION

2.4 - METAVERSE EXPANSION: HSBC CASE STUDY

HSBC is the second major bank to consider the Metaverse and has announced its intentions as a major player. While most of us are trying to reconcile what will and won't work in the Metaverse and whether "traditional" services will still be relevant, HSBC has taken a daunting step forward.

HSBC has been interested in the blockchain for years and recently bought land in the Sandbox. From the sound of its release, it is looking more at the co-creation of novel experiences, rather than skeuomorphic translations of physical offerings into a low fidelity environment.



"Through our partnership with The Sandbox we are making our foray into the Metaverse, allowing us to create innovative brand experiences for new and existing customers. We're excited to be working with our sports partners, brand ambassadors, and Animoca Brands to co-create experiences that are educational, inclusive and accessible," said Suresh Balaji, HSBC CMO, Asia-Pacific.

HSBC is also working with Animoca brands. Similar to J.P. Morgan. Instead of being against the Metaverse as a centralized intermediary, they are looking at aspects of how they can embrace it.

Sources:

1. https://www.ledgerinsights.com/hsbc-buys-nft-land-in-sandbox-metaverse-game/

2. HISTORY AND EVOLUTION

2.5 - KEY THEMES OF THE METAVERSE

1. Games-as-a-Platform – Games are redefined with an increasing amount of non-gaming social activities. This includes community events such as virtual concerts and fashion shows. Games are slowly replacing social media for enriched connections.

2. User-Generated Content – The Metaverse's decentralized nature requires community-sourced creations to scale effectively. It is expected that accessible tools will be developed for user-created experiences, games, mods, economies, and virtual worlds.

3. Cloud and Scalability – With edge computing and other cryptocurrency advancements, it is expected that there will be an increasing mass concurrency from 100 participants per shard to 10,000+. It is also expected that entry into the Metaverse will accelerate with better accessibility

4. Decentralized Economies & NFTs – There is expected to be job creation with a player-to-player economy and new virtual jobs. As a decentralized ecosystem, true and persistent ownership of unique virtual items can exist. Play-to-earn games will fuel a new economy of the century.

5. Interoperability – Technical standards, interfaces, protocols, programming, back-end, and front-end technicalities will be simplified with cross-application. Items including NFTs can be transferred across various multiverses.

6. Merging of Virtual and Physical Reality – It is highly likely real cities and objects will be replaced in a 1:1 ratio. Virtual fashion and improvement of augmentation reality are also expected

7. Avatar and Identity – Users will be able to have a persistent digital identity just like IRL to accumulate assets or improve recognition of individuals.

2.6 - WHY IS A DIGITAL IDENTITY IMPORTANT?

A digital identity is important for socialistic identity. It is important that our identity is not diluted with increasing virtual interactions, losing the human touch.

Digital identity is more than just an identity in the Metaverse, but is a digital credential, much like a passport. Beyond the expansion of customization options and the blurring of virtual and real individuals, identity is an idea of a persistent interoperable and decentralized identity.

This is referred to as the Self Sovereign Identity, which aims to protect users' privacy through DIDs (decentralized identifiers). DIDs verify identity and ownership without needing to store information on a centralized server.

One such example is Arweave or IPFS, popular for storing NFTs. The Metaverse is expected to provide a single, permissionless, decentralized identity, with self-expression beyond just the avatars of an individual. Users should be able to customize, interact with, and own assets like in real life. This includes pets, vehicles, and real estate.

3. THE METAVERSE & WEB 3.0

3.1 The Meta Web 3.0 Economies

3.2 Key Web 3.0 Metrics

3.3 Data in Web 3.0

3.4 Centralization vs Decentralization

3.5 Case Study: Pocket Network

3. THE META WEB 3.0 ECONOMIES

3.1 - THE META WEB 3.0 ECONOMIES

Web 3.0 crypto Metaverses are emerging as open virtual market economies with the development of various complex digital goods, and services, generating real world value for users.

In the early stages, worlds are built upon Layer 1 platforms with other related properties such as in-game assets, infrastructure development for developers, and many more.



The Metaverse Web 3.0 is expected to bring economic disruption, in which the ownership economy will disrupt the way work is traditionally defined.

1. Ownership is expanding beyond typical entrepreneurs and is adopted by employees. The pandemic has seen an unprecedented surge in people quitting their jobs, in what has been dubbed **"The Great Resignation"**. People are discovering their passions and learning to make a living off them rather than slaving away at a job. The Metaverse Web 3.0 is a major catalyst for this.

2. Small business ownership is also becoming a norm among digital natives or experts. In Web 2.0, there was a transition to users being able to buy directly from marketplaces. In Web 3.0, the intermediary is removed, and users can interact directly with individuals and businesses. Creating amazing branding is also easier with tools like Shopify, Orbelo, and MailChimp, which will integrate cohesively into Web 3.0.

3. Lastly, ownership is becoming a mindset. People are becoming aware that dependency on centralized platforms is risky. A major catalyst to this is creators experiencing algorithm fatigue, a phenomenon that hinders the ability to reach your audience without paid placement. This is frustrating and people are realizing that a fair, decentralized economy is needed for true ownership to shine.

In addition, it is likely that companies will become completely tokenized in the future, with users rewarded directly for various contributions.

Sources:

1. https://www.forbes.com/sites/forbesbusinesscouncil/2022/01/06/creators-and-web-30-how-economic-disruption-is-leading-to-a-new-ownership-economy/?sh=5158f9dc11f5

3. THE META WEB 3.0 ECONOMIES

3.1 - THE META WEB 3.0 ECONOMIES

Web 3.0 Economies are becoming interconnected, where decentralized protocols will interoperate in the Metaverse and provide technical infrastructure.

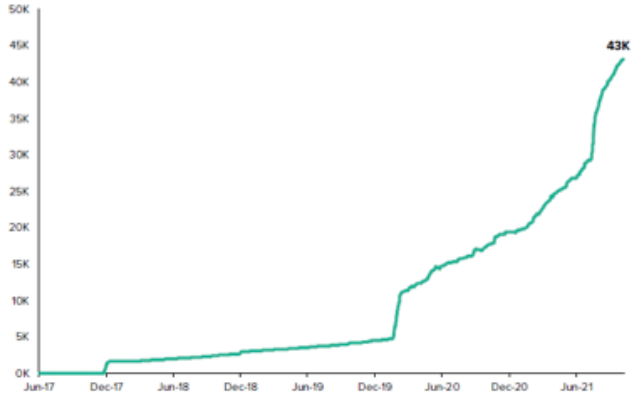
This would include areas such as:

- Payment networks like MANA built upon Ethereum.
- Decentralized Finance, enabling users to trade in-game items, using aggregators, DeFi primitives, Oracles, Data, and Units of Value
- NFT Sovereign Goods, enabling players to purchase NFTs and bring them to other virtual economies for sale or display. This would include the use of marketplaces, metadata standards, redeemable NFTs, minting houses, etc.
- Decentralized Governance, decentralized legal frameworks that take control of digital economies from centralization and allow a collective ruling of Metaverse Web 3.0 spaces. This includes areas such as DAO frameworks, Voting Mechanisms, Multisig wallets, and community wallets
- Decentralized Cloud services such as Arweave to give decentralized storage infrastructure solutions. This also includes areas such as Databases, Queries, and APIs.
- Lastly, a Self-Sovereign Digital Identity, essentially your passport, with DID's and verifiable claims via the use of smart contracts.

3. THE META WEB 3.0 ECONOMIES

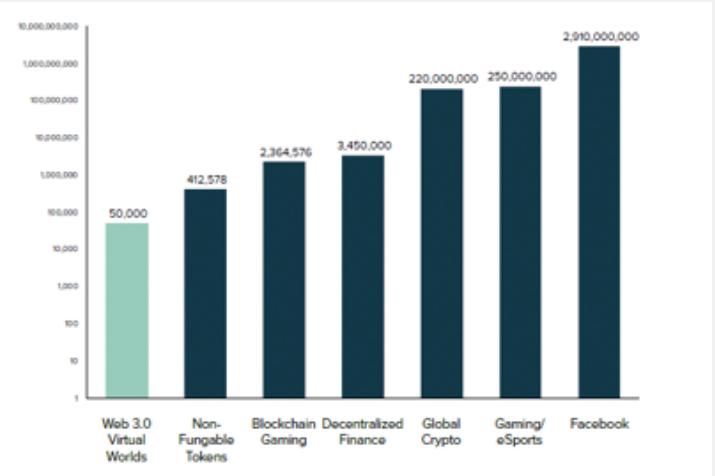
3.2 - KEY WEB 3.0 METRICS

Web 3.0 worlds and user activity has seen rapid growth over the years. According to Grayscale, the **Metaverse is a $1T opportunity** as users increase by 10x. Today, the Web 3.0 Metaverse virtual world has nearly 50,000 all-time users since 2020.



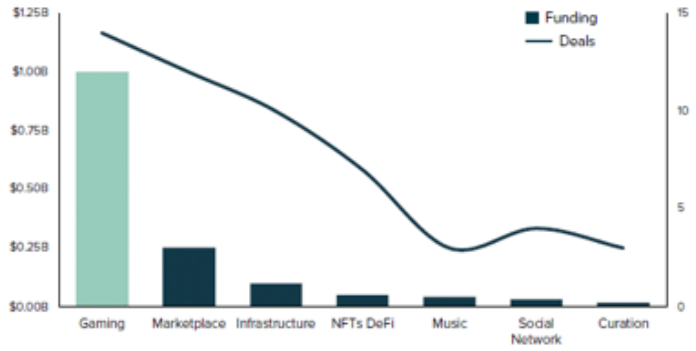
Compared to Web 3.0 segments, the Metaverse narrative is still primitive. However, if growth holds constant, the potential for it to develop into a pivotal sector remains.

It is expected that many users will convert over from traditional Web 2.0 platforms to the more decentralized internet-native Web 3.0 this decade. Currently, the global users for Web 3.0 are just a fraction of Web 2.0.



Interestingly, capital investment is starting to accelerate. Of course, Meta leads here, but the Metaverse is still primitive.

With more investments expected to follow in the coming decade, we should see improvement in its infrastructure.



Sources:

1. Dove Metrics
2. Grayscale –non-fungible, dune analytics, goldman sachs, facebook, cryptoslam
3. Taken from Grayscale – Grayscale source is non-fungible

3. THE META WEB 3.0 ECONOMIES

3.3 - DATA IN WEB 3.0

Web 3.0 is a key driver of the ownership economy. In Web 3.0 Users will not have to worry about their data and its misuse.

Web 3.0 will offer programmable interfaces with functionality for direct participation in data governance and economies. This new form of networked governance or DAOs (Decentralized Autonomous Organizations) creates mass collaboration in a fair, decentralized manner, contributing to direct democracy rather than a medium of the government for democracy.

The goal of Web 3.0 is to eliminate the middleman, making applications more efficient and secure than in Web 2.0. With DiDs, users can take ownership of their reputations from central authorities, such as credit reporting agencies. Users can choose who gets access to information based on reputation scores, verifiable on the blockchain.

The ownership economy is present and will continue to grow. Not only is there evidence of growing influence, but prominent leaders who previously did not believe in the decentralized space have begun to accept that it is not a matter of if, but when.

3.4 - CENTRALIZATION VS DECENTRALIZATION

There are both centralized and decentralized operations for the Metaverse that streamline blockchain data access. For example, blockchain infrastructure providers like Infura provide portals to blockchain data but are developed and owned by centralized entities. On the other hand, projects like Pocket Network offer decentralized solutions to accessing blockchain data.

With decentralization, some of the main concerns are how to efficiently bring data from a blockchain to applications, how to move data across blockchains (bridging) and how to do everything in a decentralized manner.

A decentralized network means that applications, nodes, and the network layer all must be decentralized.
Applications – Any software that submits API requests (queries or relays e.g. from Dune Analytics) must be routed to a public or encrypted database node
Nodes – Nodes are decentralized servers that store database indices, send session information to application, serve API requests, or even store network states like account balances.
Network Layer – The ecosystem governance, protocol roles, and economic economy. In other words, this is the ecosystem that maintains and utilizes the operations of a decentralized protocol.

Centralized blockchain infrastructure is a bit difference. While these nodes could be distributed in a decentralized network of individuals, they could be owned and operated by centralized parties.

The main difference between the two is that companies maintaining operation of nodes, can be grouped into a "company layer". In addition to providing access to full archival node data, individuals in the company can set up database indices to facilitate queries.

While centralized setups mean it is easier to develop and maintain the network, this means that there are security risks and point of failures. For example, Infura has had major outages before and complies to governmental regulations such as US sanctions. Note that Infura is also the same company as Metamask.

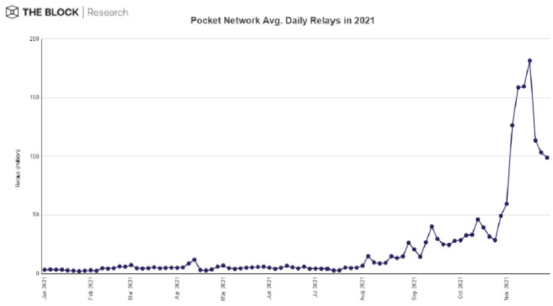
It is important to note a stark difference that even though it is centralized. Infura does not own or control the base blockchain data but act as centralized portals to access the data. If Infura goes down, users could just set their own node to access to data.

Sources:

1. https://www.forbes.com/sites/forbesbusinesscouncil/2022/01/06/creators-and-web-30-how-economic-disruption-is-leading-to-a-new-ownership-economy/?sh=5158f9dc11f5

3. THE META WEB 3.0 ECONOMIES

3.5 - CASE STUDY: POCKET NETWORK

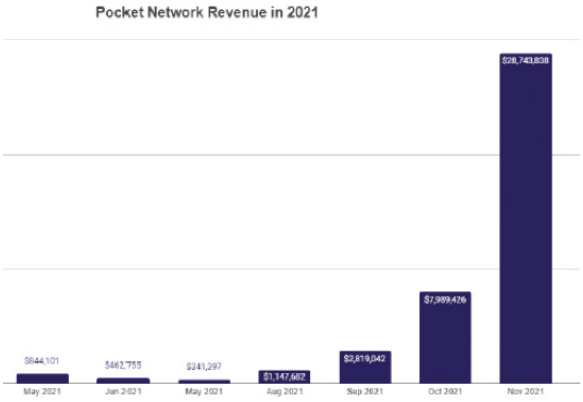


Pocket Network incentivizes participation as node providers. Recently, Pocket Network saw a surge in network activity.

Average relays have skyrocketed over time last year. Relays refer to application requests targeting any public database node.

In the Pocket Network, an application stake is the entry point to register an application in the network. Apps lock POKT within the network to receive an allocation for relay execution via network nodes, determined via decentralized governance of monetary policy and protocol roles set by the Pocket Dao. The amount of Pocket locked has been increasing significantly since 2021.

When staked tokens (locked) exhaust the relay request reserves, these are paid out as "revenue" to relay providers, Pocket DAO, and POKT block producers.



So how does Pocket Network translate relays into revenue?

Number of Relays x 0.01 x POKT Price

The chart on the right shows POKT revenue in 2021 based on average daily relays and POKT price per month.

So far, active validator nodes have increased significantly since its inception, to close to 9,000 nodes today, helping to maintain the decentralized state of its network.

Pocket Network will be a key to providing Web 3.0 infrastructure for the decentralized Metaverse. However, there is no certainty that these rewards are sustainable in the long run.

The more node providers join, the more diluted the emissions. However, with attractive rewards and increased stakers, this does represent real demand for Pocket Network. Given its integration with Harmony and other protocols, it could help with virtual network economics.

Unfortunately, decentralization is still hard due to its technical nature. Compared to its competitors, it is only handling a small number of transactions every day. Its centralized competitors, such as Infura, had over 2.4 billion Ethereum relays on a record day in 2020.

But, as users began to realize the need to completely decentralize, we could see higher utilization of solutions such as Pocket Network

Sources:

1. The block – thunderheadotc.com (price) and c0d3r.org (relay counts)

4. NFT'S CONTRIBUTION TO THE METAVERSE

4.1. Mainstream Metaverse

4.2 Mainstream Metaverse
– Korean Tourism Organization Study

4.3 Gamification of Finance

4.4 NFTs as an early building block

4.5 Metaverse Collateral

4.6 Figment : Case Study

4. NFT'S CONTRIBUTION TO THE METAVERSE

4.1 - MAINSTREAM METAVERSE

The Metaverse is going to change the way we interact. Previously, pre-pandemic cultural contents were based on face-to-face experiences. Post-pandemic, the virtual world has become a notable alternative, with similar offerings as face-to-face interactions.

With advancements like edge computing, graphic technology, and smartphones, the Metaverse has emerged as a new form of human culture.

NFTs have experienced a huge surge in interest. Once considered a small section in the cryptocurrency space, NFTs now dominate mainstream discourse as many grapple with the idea of digital property being implemented through singularity.

NFTs have become a way to signal cultural relevance for companies, who may decide to purchase popular NFTs or create their own. Some even create their own Metaverse, seen in the Korean Tourism case study.

4.2 - MAINSTREAM METAVERSE - KOREAN TOURISM ORGANIZATION STUDY





Metaverse marketing is no longer an opportunity, but a mandatory tool for the KTO (Korean Tourism Organization).

Zepeto is Korea's top Metaverse platform with over 200 million users, of which 80% are below the age of 30. Notably, 90% of their users are from overseas.

Recognizing the changing digital economy for the younger generation, KTO introduced a new Metaverse map comprised of unique charms from 5 Korean cities including Busan, Mokpo, Andong, Gangneung, and Jeonju.

The organization has even offered their own version of Squid Game using the Metaverse platform Roblox. Avatars can play popular Korean games, take pictures of attractions, and even "taste" some of Gangneung's specialties, such as rice cake or tofu ice cream.

Virtual tourism is not just welcomed by international tourists but also domestic tourists. Local Zepeto users in Korea were ecstatic that they were able to experience a new way to spend time with their family and friends without having to wear masks.

"I saw many celebrities eating instant ramen and kimbap on a bench at Han River parks. It was so cool to do that with my friends in Zepeto," a middle school student surnamed Park from Changwon, South Gyeongsang Province, told The Korea Herald

It doesn't just end here. The Metaverse has been credited in elevating the K-pop Industry to a new level. In Korea, culture, sports and tourism is promoted via Metaverse concerts featuring individuals such as BoA, girl group asepa, rapper Wonstein and many more.

The ever-growing concept of the Metaverse is not just limited to tourism and music sectors for Korea. Constantly under development, it dabbles in areas including future, food, and entertainment.

Sources:

1. http://www.koreaherald.com/view.php?ud=20220103000633

4. NFT'S CONTRIBUTION TO THE METAVERSE

4.3 - GAMIFICATION OF FINANCE

Gamification narratives are still in their nascent stages, but the possibilities are endless. Gen Z today is showing an increasing interest in becoming financially literate. As the lines between gaming and financial instruments blur, corporations are taking advantage of this as an enterprise use case for brand marketing services to more tech-savvy individuals.

A rise in institutional interest will only skyrocket as opportunities pique corporate interests in areas like the supply chain, real estate, and healthcare.

Neobanks have been offering an innovative range of products ranging from managing personal finances to platforms for financial education.

It is expected in the future that fans and communities can share the financial success of their favorite products directly using this Gamification.

4.4 - NFTS AS AN EARLY BUILDING BLOCK

NFTs can serve as a building block in the Metaverse for many applications, such as the "TradArt" world. This area is plagued by middlemen and companies who do not compensate creators enough for their contributions.

For example, the digital artist Beeple only experienced moderate success in his career but experienced a complete turnaround with NFTs. After a painstaking 14 years of producing a collection he called "Everydays", he sold the first 5000 "days" of his collection at Christie's for $69.4 million.

NFTs will leverage the new decentralized, fair, and open economy that will help users, creators, and artists interact directly. Opportunities include finding a willing buyer and seller without an intermediary.

NFTs, alongside the Metaverse, will not be driven by a single purpose application, but instead an entire spectrum of online activities. NFTs will be a primary catalyst, and companies such as Tiktok, Twitter, Microsoft, and Disney have all announced their own plans to get involved at this intersection of crypto.

With such fervor in these areas, it is likely NFTs will be instrumental in contributing to the development of these technologies throughout the next decade.

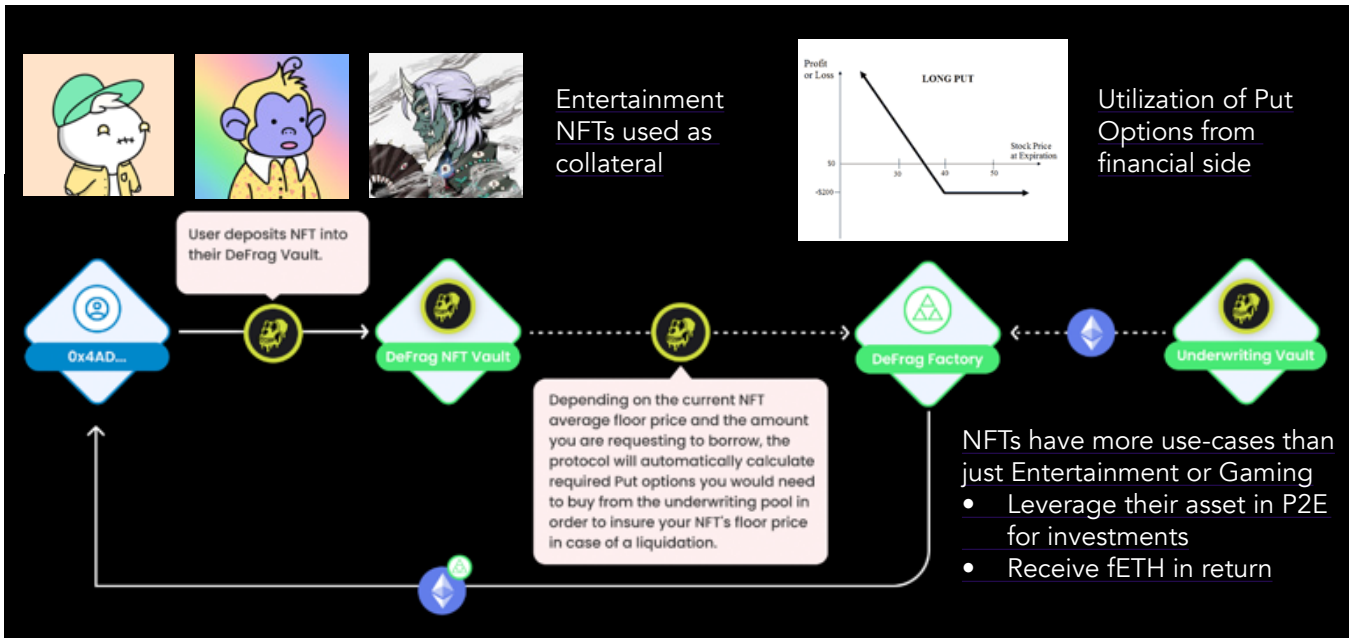


4. NFT'S CONTRIBUTION TO THE METAVERSE

4.5 - METAVERSE COLLATERAL

It is important to understand that NFTs can now be considered an asset in a basket of goods, that can be borrowed and lent against. While NFTs are less liquid, collections like BAYC will mean that MetaFi will soon be common (more on that later).

Furthermore, it is becoming increasingly interesting that the interplay of fungible and non-fungible social tokens is being generated for and by the creator class, as an opportunity to remove the need for intermediaries. This allows people to have a stake in a creator or community franchise and participate in creating future value.



As entertainment and gaming mature, especially for P2E economies, it is only natural that NFTs from these sectors can be used for financial purposes. One such example is Defrag protocol, allowing users to utilize their entertainment assets as liquid collateral.

Users will be able to stake their NFTs as collateral via the use of put options, which allows their idle assets to be put to good use by leveraging it and receiving fETH in return. Users can reinvest this fETH to make a profit elsewhere, before they repay their loan. In other words, this provides financial opportunities just like DeFi, and this is just one aspect of financialization, with many more to come.

4. NFT'S CONTRIBUTION TO THE METAVERSE

4.6 - FIGMENT : CASE STUDY

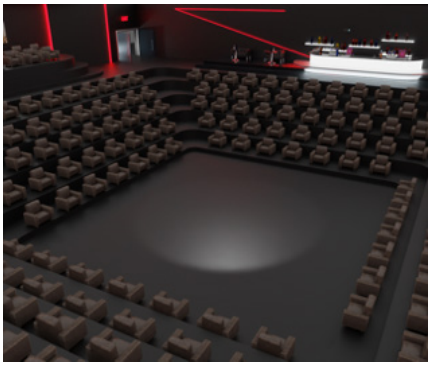
It is important to understand that NFTs can now be considered an asset in a basket of goods, that can be borrowed and lent against. While NFTs are less liquid, collections like BAYC will mean that MetaFi will soon be common (more on that later).

Furthermore, it is becoming increasingly interesting that the interplay of fungible and non-fungible social tokens is being generated for and by the creator class, as an opportunity to remove the need for intermediaries. This allows people to have a stake in a creator or community franchise and participate in creating future value.

Figment currently has prime notable areas, of both an exclusive virtual club and physical club in countries around the world. The cloud country club allows users to connect anytime and anywhere, while the IRL houses are boutique residences that allow citizens to eat and drink alongside other global citizens and browse through their dynamic NFT Museum.

The pictures on the right are 3D Rendered Models of the cloud country club.









2022 will see the launch of their NFT museum and exclusive drops of Figment X tokenized physical merchandise, in addition to Residence Retreats.

As an unprecedented organization, all of Figment NFT Sales and royalties net of costs are funneled back into the development of the country club.

Eventually, they plan to move into interest areas such as Metaverse Condo and Land Sales, and expanding to other countries in 2023.

5. METAFI - DEFI FOR THE METAVERSE

5.1 The Current Economy

5.2 Defining MetaFi

5.3 Key Catalysts for MetaFi

5.4 A framework for MetaFi

5.5 MetaFi Key Metrics and Activities

5.6 Limitations Today

5. METAFI - DEFI FOR THE METAVERSE

5.1 - THE CURRENT ECONOMY

Today, there are billions of dollars trapped in existing Web 2.0 platforms such as social media platforms like Meta, Instagram, or TikTok. Web 2.0 Can be referred to actively and deliberately built "moats" to extract as much money as possible for the company and its shareholders.

Web 2.0 economies operate on the principle of shareholder supremacy at the expense of the user, and such is the reality of centralized platforms. We can agree that the economy today is not considered part of the Metaverse and is structured around profit maximization. Currently, assets are not freely tradable across platforms and are primarily locked in the value of the platform equity alone.

Currently, there are many limitations with this type of economy. These include Limited Inclusion, Dynamic and Subjective Terms & Conditions, and Siloed by Design.

1. **Limited Inclusion** – Creators in today's economy are controlled by centralized companies or institutions who act as the middleman and take huge cuts from creator earnings. **In short, the current economy favors institutions and companies looking to gain maximum profits from middle manning a service.**

2. **Dynamic and Subjective Terms & Conditions** – Participants in the current economy cannot trust centralized services that have the right to modify terms and conditions without notice. **In short, the rules of participation in these platforms are not fair, clear, or creator-friendly. These platforms are not auditable and preexisting conditions can change at any time.**

3. **Siloed by Design** – Due to unconventional design, users and creators cannot transfer value off the platform or exit the closed economy. This also means that over time, a monopoly becomes established.

5.2 - DEFINING METAFI

So, what is MetaFi? Similar to DeFi (Decentralized Finance), MetaFi is a term for Metaverse Finance inclusive of the intermingling of fungible and non-fungible tokens. It includes protocols, products, and services.

MetaFi utilizes two key core principles from DeFi.

1. It is unstoppable and cannot be controlled by government regulation

2. It utilizes composable infrastructure, acting as a money lego for developers

In a nutshell, MetaFi borrows DeFi principles and widens the Metaverse by utilizing a mixture of tokens, alongside new organizational structures of community governance like DAOs.

As our already world becomes more digitized, billions of users will flock to MetaFi ecosystems.

5. METAFI - DEFI FOR THE METAVERSE

5.3 - KEY CATALYSTS FOR METAFI

1. Maturity of DAO Governance Models – Maturing DAO stacks allow for collective governance of purely on-chain digital and financial service provisions without the need for intermediaries like banks or institutions. These plat a vital role in MetaFi adoption and introduce prospects for community-based insurance to users.

2. Development of Financial Stacks - Previously, the complexity of DeFi stacks has driven developers away. However, through NFT platforms, creators and communities can establish the economic terms for creative exchange through things like perpetual royalties for creators through social tokens. It is unstoppable and cannot be controlled by government regulation

3. Gamification of Finance – Probably the most important, the evolution of play-to-earn games and new ways of monetizing data help draw Gen Z closer to MetaFi. Gen Z has shown an increasing interest in becoming financially literate as compared to previous generations. This has lead to institutional interest such as neobanks arranging interesting ways for individuals to arrange personal finances. On top of this, the lines between games and financial instruments become increasing blurred, such as institutions like J.P. Morgan and HSBC entering the Metaverse.

5.4 - A FRAMEWORK FOR METAFI

The framework for MetaFi is a structure of components, just like how Web 3.0 follows a framework of D.O.V.E, MetaFi follows a framework of F.D.V.

1. **Foundations** – This refers to the building block of any structure. Foundations include frameworks or protocols such as layer zero/one/two solutions. These frameworks allow applications to be built on top of them. If a Metaverse application does not integrate with this core layer, its economic and creative value will stagnate and fade away. We can see similarities in the traditional world, where services that do not adapt and integrate with the wider ecosystem fade due to being irrelevance.and HSBC entering the Metaverse.

2. **DeFi** – The building block, consisting of small financial applications available on core protocols. For example, the Fantom ecosystem is a complex and diverse ecosystem that can be referred to as a "money lego" haven and enables complex financial dynamics through smart contracts. While the Metaverse narrative isn't strong on the Fantom ecosystem, it could be in the future.

3. **Verses** – The final component in MetaFi, is a collection of domains or parallel verses which build the Metaverse. The different virtual worlds connect with the foundational layers of verses according to compatibility and cost-effectiveness. This is still in its nascent stages and incredibly dynamic, with MetaFi being a phenomenon taking place at the intersection of fungibility and non-fungibility.

Sources:
1. https://101blockchains.com/metafi/

5. METAFI - DEFI FOR THE METAVERSE

5.5 - METAFI KEY METRICS AND ACTIVITY

MetaFi has had several developments in many key areas. Its respective definitions will be subject to change as it evolves. It is important to note that development in these areas is enabled by DeFi applications, data oracles, or various foundational layers mentioned previously. Some notable developments include:

1. Virtual Worlds
2. Games
3. Avatars
4. Marketplaces
5. Access Tokens
6. Yield-Bearing NFTs

Virtual Worlds – Contrary to popular belief, virtual worlds may or may not mimic the real world and its physics. If they do, and they most likely will, it will take the form of NFTs for scarcity. The adoption of virtual worlds has moved significantly since its close linkage and inception of in-game currencies and governance tokens.

Games – Traditional games are primarily for amusement. What sets Metaverse gaming apart are the play-to-earn element and many other use cases. Blockchain gaming has seen significant adoption since its inception at the start. Almost half of all active wallets have connected to a game, with the top 10 blockchain games collectively boasting 4m monthly users.

Avatars – These are designed to create unique identities for users, including 3D Avatars usable in Metaverse spaces. Popular projects would include CyberKongz or Supducks, that distribute native tokens to their NFTs.

Marketplaces – This is a key metric as marketplaces are digital places that help match supply and demand. These include OpenSea, Rarible, and many others (see NFT Report). From there, NFTs can be used as financial assets through fractionalization, which is a way to obtain liquidity via fungible tokens with proportional ownership. This results in an index fund solution for the given category of the NFT.

Access Tokens – Can come in both fungible and non-fungible forms, which grant access to a community, specific "alpha", or maybe even future benefits.

Yield-bearing NFTs – NFTs can produce yield indirectly or directly. For example, NFTs can be used as collateral to take out loans. This is still in its experimental stages as NFTs are highly volatile and illiquid. NFTs may also grant access to a native token, creating a social token economy. For example, some tokens allow users to participate in raffles to win blue-chip NFTs.

5.6 - LIMITATIONS TODAY

For MetaFi to reach its true potential, a lot of development still needs to happen. On the consumer side of things, there are still several limitations that must be overcome. For example, NFT appraisals to identify how much a collection is worth. Furthermore, there is a need for standards across blockchains for NFT interoperability.

In addition, to fully unlock DeFi for the Metaverse, NFTs need to be easily compatible with DeFi protocols. There are currently 3 ways this can be solved.

1. Fractionalization of NFTs – This has already happened, but the exact concept is still being ironed out. Notable projects fractionalizing NFTs include Fractional or DAOfi.
2. NFT-isation of DeFi – This refers to upgrading existing standards of DeFi protocols so that they can accept NFTs as a form of collateral. This is a little bit tricky since NFTs are even more difficult to value thanks to their non-fungibility.
3. NFTs as derivatives – This refers to creating liquid digital assets whose value is dependent on the value of off-chain assets or in-game items, or even linked to data oracles. For example, artists could create art and create an NFT derivative to use as collateral to mint synthetic assets.

6. POSSIBILITIES IN THE METAVERSE

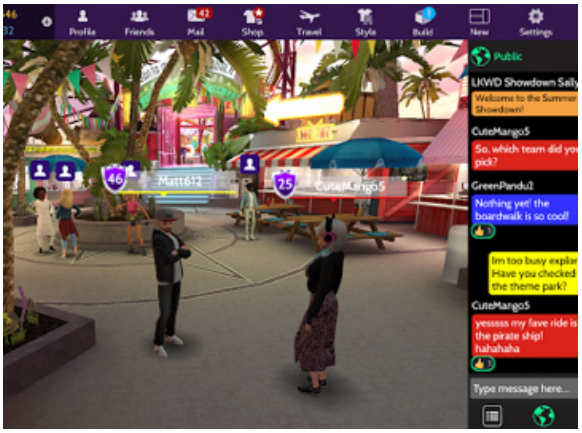
6.1 Unlimited Creator Opportunities
6.2 Key Consumer Perspectives and Insights
6.3 The Rise of Gaming Guilds
6.4 Fund-raising Opportunities and Activity
6.5 New Primitives for Games – Loot Case Study
6.6 Multichain Solutions

6. POSSIBILITIES IN THE METAVERSE

6.1 - UNLIMITED CREATOR OPPORTUNITIES

According to Epic Games, the Travis Scott Astronomical Fortnite concert drew 28 million unique players who collectively attended the event 46 million times, meaning almost all players attended more than once.

Roblox hosted a Lil Nas X Concert, garnering over 33 million unique players. The CFO of Roblox, Mike Guthrie, stated that not only did the usage of ages 13 and older double after the concert, but the demographic for those aged 13 and older also account for 44% of the Daily Active Users.





Games offer unlimited creative possibilities. For example, Avakin Life & Zepeto are key mobile apps for brands, pioneering the mobile Metaverse space for creator opportunities.

Avakin Life is a social simulation platform with over 200 million downloads and 1.3 million daily active users. They develop all their assets and worlds, unlike Metaverse platforms like Roblox.

Zepeto is a creator-powered social app, which reports more than 150million registered users.

Just like Roblox, it follows a concept where 3D virtual worlds and items are powered by creators. In addition, they promote Social Finance and utilize social-focused features such as allowing users to create animations using their avatars.

Both Individuals and Corporations can tap into this creator economy, evident by their partnership with famous brands such as Gucci.





6. POSSIBILITIES IN THE METAVERSE

6.2 - KEY CONSUMER PERSPECTIVES AND INSIGHTS

To get an understanding of potential Metaverse engagement, NewZoo conducted around 5,500 surveys spread evenly over four key markets.

They asked the participants what they would want to do to get involved in the Metaverse and whether it would impact future participation.

A significant finding was that users were **unsure of what the Metaverse really meant**. Instead, they described it as a being related to using game and game-like worlds for socializing and as a creative platform, where activities and assets were presented clearly just like real-life activities and transactions.

Some other key points include consumers being enthused by the idea of the Metaverse and generally positive about it. There is currently quite a strong appetite to get involved.

However, it is noted that the enthusiasm varies per market. It is generally noted that out of the 4 countries surveyed by NewZoo, China and U.S were generally more positive as compared to the UK and Japan.

It is also highly likely that the Metaverse will encourage consumers to play even more games. Most consumers expect that jumping into the Metaverse will not just increase total game time, but also attract a significant portion of non-gamers.

Lastly, the Metaverse is expected to be evolutionary and not revolutionary. In the future, the Metaverse may even replicate our physical reality on a 1:1 scale.

6.3 - THE RISE OF GAMING GUILDS

The ability of Play-to-earn games has opened up various new business models and opportunities. A particular interesting category is gaming guilds. Guilds support players from areas such as onboarding to lending in-game assets, taking a commission in return.

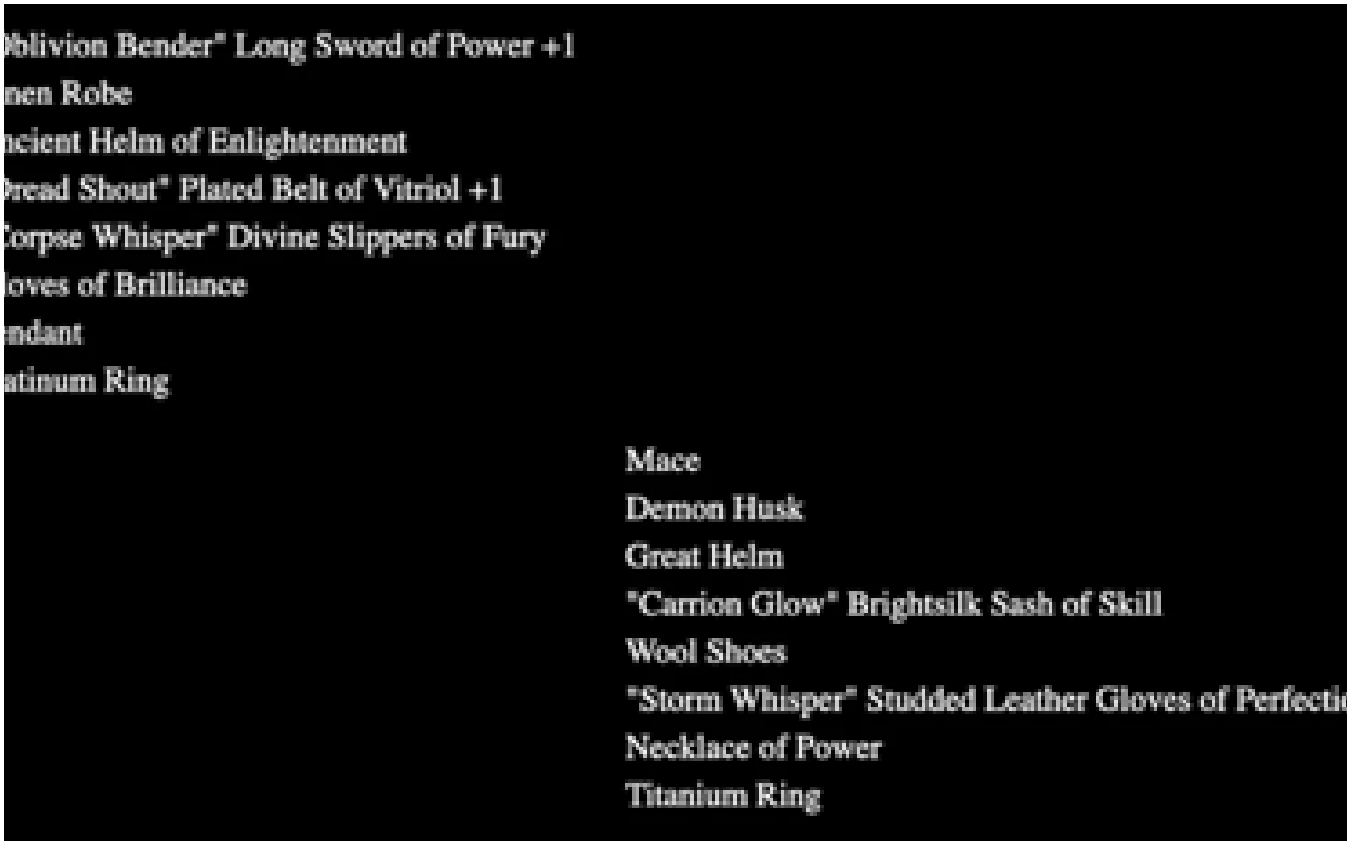
Most of these guilds are structured as Decentralized Autonomous Organizations. Guilds are essential as when the popularity of games increase, in-game assets necessary for playing the game become more expensive to own. This makes the barrier to entry extremely high for many players. In this case guilds buy them for players and lend it to them. In return, they take a small portion of the players earnings. Some examples include Yield Guild Games, Merit Circle, Avacado Guild, AAG Ventures and many more. Lastly, the Metaverse is expected to be evolutionary and not revolutionary. In the future, the Metaverse may even replicate our physical reality on a 1:1 scale.

6.4 - FUND-RAISING OPPORTUNITIES AND ACITIVITES

2021 was a year for many fundraising opportunities. The Metaverse sector itself raised nearly $28 billion, not including IDOs.

Besides just companies building in this space, crypto-native projects have also raised funds through different methods such as Venture Capital, ICOs and IEOs (Initial Coin/Exchange Offerings). These offerings refer to where new projects use existing exchanges as distribution platforms for Metaverse-related sales. (both NFT and token sales)

6. POSSIBILITIES IN THE METAVERSE



6.6 - NEW PRIMITIVES FOR GAMES

Amidst the speculation of the NFT frenzy, and soon to be the Metaverse frenzy, experimentation with unorthodox modalities flourished, which lead to the development of new primitives for games.

One such example is Loot. Loot came up as a prime example of the untapped potential of NFTs in the Metaverse. Loot is a set of randomized adventurer gear, generated and stored on-chain.

Instead of relying on the conventional top-down approach, it was designed to pioneer a bottom-up approach. Collectors would only be able to mint the base layer of the NFT and the community can then construct the context and front end around it. This can also be considered an NFT Lego. Shortly after Loot's launch, the community launched AGLD, which is used as an in-game currency.

In short, Loot can be considered a composable "DNA", inserted into games, and the visual representation of the adventurer's gear and additional features are based on this DNA code.

6. POSSIBILITIES IN THE METAVERSE

6.7 - MULTICHAIN SOLUTION

The Mass Adoption of the Metaverse and NFTs will most likely be driven by multi-chain scaling solutions. If the aim is for the broad adoption of Web 3.0, cheaper transactions on a decentralized network are essential, especially on networks like Ethereum. This has resulted in many Layer 2 Scaling Solutions coming out such as Optimism, Arbitrum, and ImmutableX launching on top of Ethereum.

That being said, there are also other layer-1 scaling solutions that users have investigated for the Metaverse, such as Solana, Harmony, Fantom, Algorand, and Avalanche. NFT and Metaverse activity has already spread beyond Ethereum and will flourish on other chains. Hence, solutions that allow interoperability will be crucial to drive mass adoption.



7. THE FUTURE AND CLOSING THOUGHTS

7.1 Bifurcation of MetaFi
7.2 Securing Skilled Talent
7.3 Creation of More Innovative Experiences
7.4 The Metaverse as a Key Factor for Companies
7.5 Privacy and Ethics
7.6 Accessibility and Scale
7.7 Author's Closing Thoughts
7.8 Community Discussions

7. THE FUTURE AND CLOSING THOUGHTS

7.1 - BIFURCATION OF METAFI

A similar debate would be the ultimate degree of (de)centralization for DeFi.

Institutions are eager to deploy capital into the Metaverse but hesitant due to regulatory uncertainties. Some believe that the development of centralized entities controlling the Metaverse defeats the purpose of decentralization.

However, trustless solutions may not be the end all be all. For example, some applications require trust, such as borrowers and facilitators of uncollateralized lending. Open Finance perhaps should include a balance of decentralization and transparency.

Many decentralized applications nowadays actually rely on centralized components and will be forced to choose sides as regulation ramps up. It is highly likely that a compromise will be reached between the two, and both centralized, as well as decentralized entities, will have their own sections of the Metaverse.

Either way, the bifurcation of MetaFi is inevitable and we can only speculate on the degree to which this will happen.

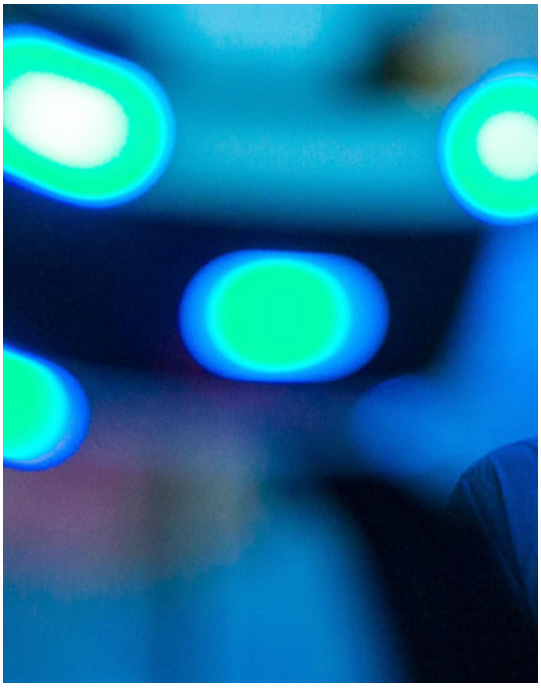
7.2 - SECURING SKILLED TALENT

As the Metaverse progresses, the need for skilled individuals increases as well. It is estimated that the demand for Web 3.0 hires to available talent is currently 14:1.

Companies like Meta, Apple, and Microsoft have been aggressively recruiting and poaching talent.

Nvidia is also looking for skilled omniverse engineers. Their products will be imperative to power the Metaverse, Graphics, and Animations.

7.3 - CREATION OF MORE INNOVATIVE EXPERIENCES



Once companies have sufficient talent and capital to develop the Metaverse, many innovative experiences will be developed. The future of work will be defined by the Metaverse and Web 3.0 infrastructures like Decentralized Autonomous Organizations.

Not just that, but school and social life too. Educational videogames or even community-based experiences will be developed in the Metaverse, allowing participants to gather for live virtual events.

Furthermore, we would likely see a new industry being created. Instead of e-commerce we will now have e-apparel. Companies have long capitalized on the sale of digital items and the Metaverse will only advance it.

For example, companies such as Nike have begun selling virtual shoes and accessories, and luxury brands like Gucci and Luis Vuitton are selling digital clothes. Modelling our exact reality, Meta-malls are beginning to pop up, allowing individuals to show up in VR stores to shop for both physical and virtual outfits.

Sources:

1. https://www.forbes.com/sites/hannahmayer/2022/01/24/the-future-of-the-metaverse-what-2022-has-in-store-for-the-immersive-digital-world/?sh=516107d6335a

7. THE FUTURE AND CLOSING THOUGHTS

7.4 -THE METAVERSE AS A KEY FACTOR FOR COMPANIES

The Metaverse acts as a massively scaled and interoperable network of rendered virtual worlds that model our very reality, so any company not participating in this space will lose out against competitors eventually.

Companies that argue that the Metaverse is just defined by gaming are wrong, as it further develops to include a stream of sectors.

7.5 - PRIVACY AND ETHICS

With such great Metaverse activity tracking, it has the potential to become far more powerful than normal web tracking. It is essential that we work toward an open Metaverse to prevent it from being controlled by select entities.

Organizations such as Crucible are working on this.

Beyond just the politics of privacy, moderation of the Metaverse on a global scale is a significant challenge. In today's society, even governments cannot fully regulate the internet, resulting in the creation of the Deep and the Dark Web. It is easy to imagine how the Metaverse could evolve to include challenges like disinformation, harassment, and identity theft.

7.6 - ACCESSIBILITY AND SCALE

In terms of future Metaverse prospects, the Metaverse is still not accessible or interoperable at a large scale, putting emphasis on multi-chain solutions.

Currently, new protocols and standards are currently being developed and many more have yet to come. These new standards will hopefully be able to create an open Metaverse for all with equal participation, highly likely to be integrated and driven with Mobile devices.

Mobile is one of the greatest global ecosystem system marvels today, and it is quite difficult for the Metaverse to function without it.

7. THE FUTURE AND CLOSING THOUGHTS

7.7 -AUTHOR'S CLOSING THOUGHTS

The Metaverse will be a radical change to the way we live today. In particular, I am keen on Web 3.0 models complementing the Metaverse narrative itself as a fresh take on current governmental systems.

For example, DAOs are expected to change our workplace infrastructure today, and help reduce and prevent coordination failures such as the prisoner's dilemma. Decentralization means that obliging to a canonical company decision can be vociferously debated if decisions are unilateral.

The Covid-19 pandemic has acted as a catalyst for change in our current world. It has also sparked workers to rethink their careers, work conditions, and long-term goals with the "Great Resignation".

It will be interesting to see how Metaverse developments move in tandem with Web 3.0 and DeFi to form a cohesive and new robust ecosystem. The very birth of cryptocurrencies was not created to get you rich or just develop the Metaverse, it was to set you free.

7.8 - COMMUNITY DISCUSSION @ASTURIIAS

My hope is that it is used for good instead of for harm, with opportunities to interact and have multiple lives. To meet and get to know people, building relationships that we otherwise would not. To live in the way that we would want, with fewer limitations.

My hope is not a strong one. We already see the hand of large corporations and crypto whales in this, committing huge amounts of money into what is essentially advertising space. Certain things are massively overpriced or inflated as status symbols or Veblen Goods. We are taking what is sickening about our world and putting it into the Metaverse.

Ads and targeted news already scream for our attention in Web 2, on our phone and computer screens. In a more immersive future, we need to take back Web 3 and truly make it for the common people.

Crypto has gathered quite a large proportion of very greedy people. It also has gathered a large proportion of very helpful people who share and enrich each other. I want our future Metaverse to be dominated by the latter, not the former. - @Asturiias

7. THE FUTURE AND CLOSING THOUGHTS

7.8 - COMMUNITY DISCUSSION @EGGBOMB

The future of Metaverse is still unknown. Too many factors to matter, too many influence to sway. It does not provide equal opportunity; it does not grant freedom as there is no true equality or freedom in this world. But it gives you the chance to liberate yourself.

Metaverse brings about a new dimension which bridges across internationally on Web3, it breaks boundaries of what we presume to be First or Third World country because the Metaverse simply does not care. It will take some time for it to be mainstream but when it happens, it's a takeover. There is no boundary. -@Eggbomb

7.8 - COMMUNITY DISCUSSION @LEONARD

The Metaverse will be the place where humanity can express itself, unbound by the laws of physics, only limited by imagination and complexity of computer code.

I look forward to new forms and social games that will bring forth a renaissance in community and aesthetics. -@Leonard

7.8 - COMMUNITY DISCUSSION @JONW3INER

One of the trends in the cryptocurrency at the moment is the rise of Metaverse. With mainstream attention put on Meta Platforms and their huge planned expenditure on creating their version of the Metaverse, what does it mean for the offerings in cryptocurrency?

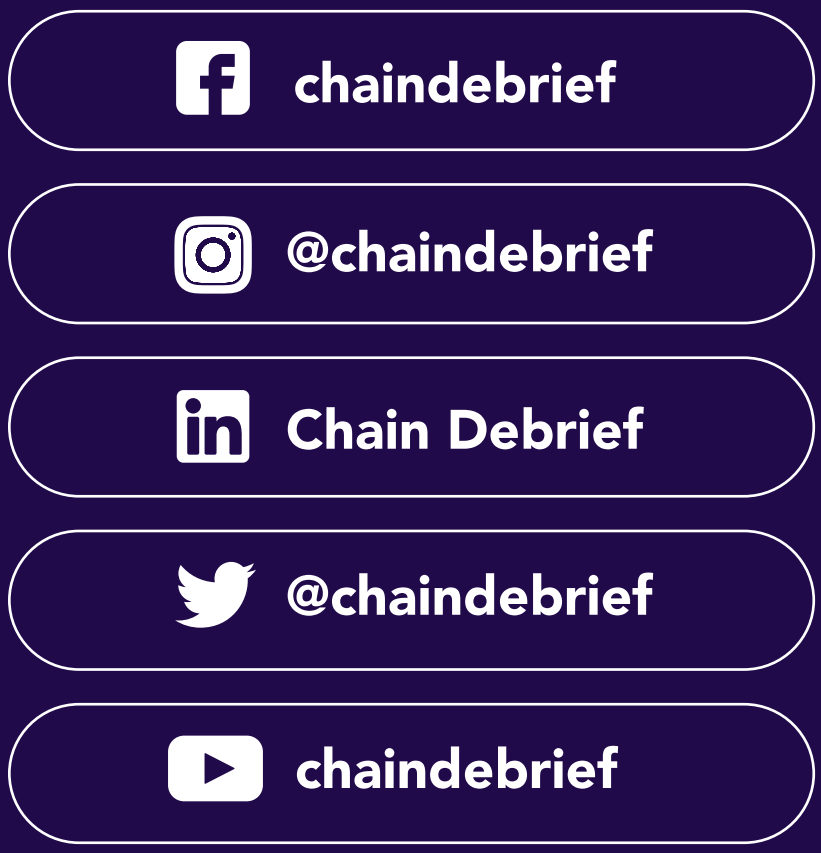
A lot of focus has been put on the Metaverse (and the utility that it brings), due to the Meta Platforms and its drive to create it's own Metaverse. But what actually is a Metaverse? Is it a space, or the medium in which people communicate in/through? Or is it a a platform in which people use to push information and service out, just like how the current form of the internet is?

Arguably one can say that we are already living in an age of Metaverses - just not in the specific sense of "Ready Player One". Looking at that movie, you can also see that decentralisation was not a part of that universe, where one organisation has full control over the game, influencing how the game works.

When you think crypto-based Metaverses, your attention goes to Decentraland, The Sandbox. These platforms have the lions share of current offerings of the Metaverse, and has some mainstream adoption (J.P. Morgan with their Onyx Lounge in Decentraland, Snoop Dogg driving more people into Sandbox with his purchase of land there). Platforms like these are also creating plenty of job opportunities within them, from casino dealers running casinos in Decentraland, to designers who create the world that people can interact with, in both Decentraland and Sandbox.

There are plenty more opportunities lying within these currently available Metaverses, and more are being created as we speak. Ultimately Metaverses and how they will be used, will depend heavily on how the company executes on their vision of the Metaverse. Jobs may be created, and fortunes could be made by the first movers in this ever growing space.-@j0nw3iner

EVERYTHING YOU NEED TO KNOW ABOUT CRYPTO



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